V. From Bornological to Coarse Cohomology

0. Reminder

- $\mathbf{TopBorn}$: $\begin{cases} (X,\tau,\mathcal{B}) & \text{topological bornological space} \\ f: X \to Y & \text{cont. proper} \end{cases}$

- $CB^*: \mathbf{TopBorn}^{op} \longrightarrow \mathbf{Ch}$

  $X \mapsto CB^*(X;G)$

  $CB^n(X;G) = \{\varphi: X^{n+1} \to G \mid \Delta_n \cap \overline{\mathrm{supp}\,\varphi} \text{ bded}\}$.

  $\underbrace{HB^*}: \mathbf{TopBorn}^{op} \longrightarrow \mathbf{Ch}$

- **Thm** (R. 2025): For any $X \in \mathbf{TopBorn}$,

  $$HB^*(X;G) \cong \operatorname{colim}_{B \in \mathcal{B}} \overline{H}^{*-1}(X \setminus B; G)$$

- **Examples**: $\mathbb{N} \cong \mathbb{Z}$ in $\mathbf{TopBorn}$;

  $HB^*(X;G) = 0$ for $* \neq 1$

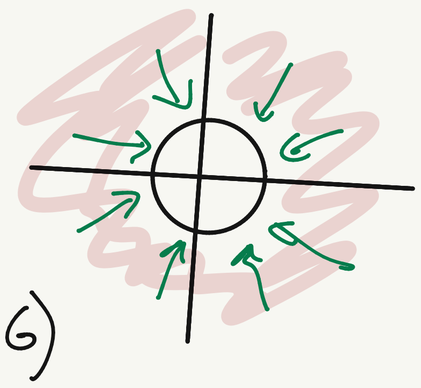
  Examples: $X = [0,\infty)$.

  For each $n \in \mathbb{N}$, $B_n = [0,n]$; $(B_n)_{n\in\mathbb{N}}$

  $$HB^*(X;G) \cong \operatorname*{colim}_{n\to\infty} \overline{H}^{*-1}((n,\infty), G) \cong$$

  $$\cong \overline{H}^{*-1}(\{*\}; G) = 0$$

  b) $X = \mathbb{R}^n$; $n \geq 1$;

  $B_n = B(0;n)$;

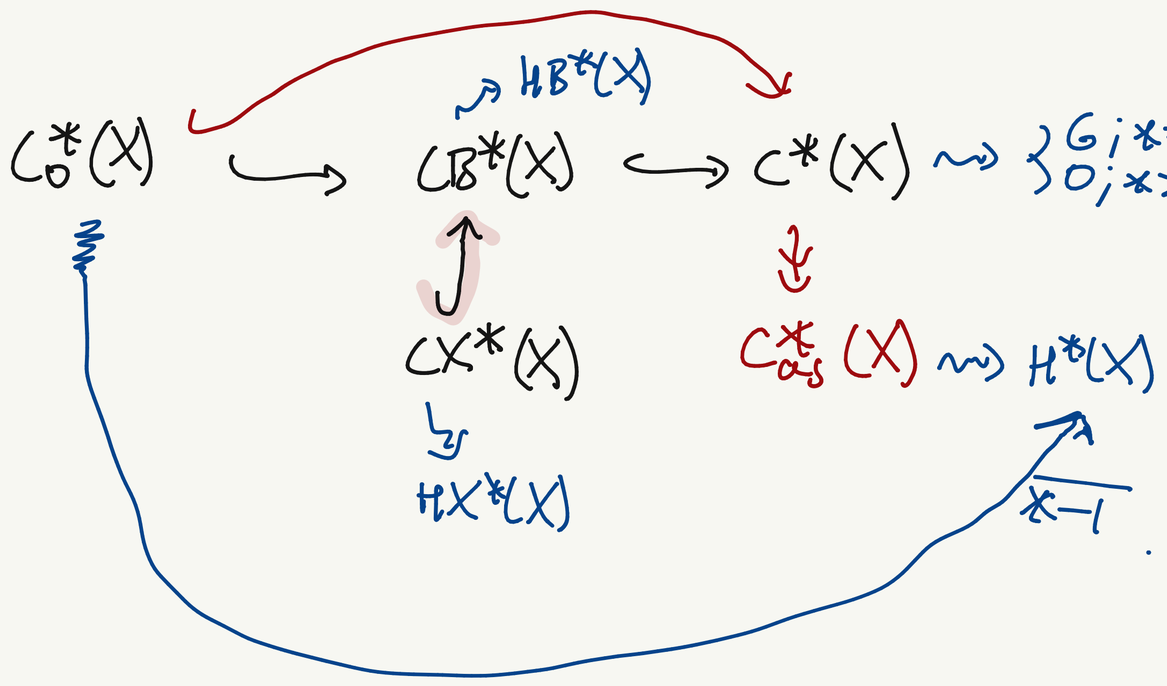
  $$HB^*(X;G) \cong \operatorname{colim} \overline{H}^{*-1}(\underbrace{\mathbb{R}^n \setminus B(0,n)}_{S^{n-1}}; G)$$

  $$\cong \overline{H}^{*-1}(S^{n-1}; G) = \begin{cases} G; & * = n \\ 0; & \text{otherwise} \end{cases}$$

  $\mathbb{R}^n \cong \mathbb{R}^m \Rightarrow HB^*(\mathbb{R}^n) \cong HB^*(\mathbb{R}^m) \Rightarrow n = m$.

Coarse vs bornological cohomology

- Assume (X,d) is a metric space.
  - ↳ Coarse space w/ metric str. $\rightsquigarrow HX^*(X)$
  - ↳ top. born. space $\rightsquigarrow HB^*(X)$.



Since $CX^*(X) \hookrightarrow CB^*(X)$; there is an induced map $HX^*(X) \xrightarrow{\Sigma} HB^*(X)$. When is it an iso?

- $\mathbb{N} \sim [0,\infty)$ $\not\simeq \mathbb{Z}$ c.e. in Coarse

  $\mathbb{N} \not\simeq [0,\infty)$ in TopBorn

  $\mathbb{Z}$

- Def'n: X is "uniformly contractible at infinity" if $\exists C, \mu: [0,\infty) \to [0,\infty)$ and a basepoint p s.t.:

$$\left[B \subseteq X,\ \operatorname{diam} B = r \text{ and } d(p,B) \geq \mu(r)\right] \Rightarrow \left[\begin{array}{l} B \text{ is contractible inside a set of} \\ \text{diameter } \leq Cr \end{array}\right]$$

$$(\exists B';\ B \hookrightarrow B').$$

* Thm: (Banerjee, 2024): If X is "uniformly contractible at infinity", then

$\Xi : HX^*(X) \xrightarrow{\cong} HB^*(X)$ is an isomorphism.

Consequence: $HX^*(X) \cong \varinjlim_{B \in \mathcal{B}} \overline{H}^{*-1}(X \setminus B)$

If $p \in X$ is a basepoint, then

$$HX^*(X) \cong \varinjlim_{r \to \infty} \overline{H}^{*-1}(X \setminus B(p,r))$$

* Examples:

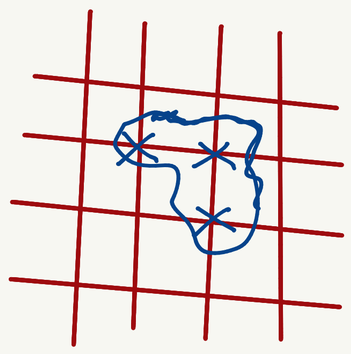
a) $\mathbb{R}^n$ is unif. contr. at infinity;

$$HX^*(\mathbb{R}^n) \cong HB^*(\mathbb{R}^n) \cong \begin{cases} G; * = n \\ 0; \text{otherwise} \end{cases}$$

b) $\mathbb{Z}^n$ is not unif. contr. at infinity;

$\mathbb{Z}^n \underset{c.e.}{\sim} \mathbb{R}^n$;

$$H X^*(\mathbb{Z}^n) \cong \begin{cases} G; * = n \\ 0; \text{otherwise} \end{cases}$$



Consequence: $\mathbb{Z}^n \underset{c.e.}{\sim} \mathbb{Z}^m \Rightarrow n = m$

Classification of abelian f.g. groups:

If A is abelian f.g., then $A = \mathbb{Z}^r \oplus T$, where T is a torsion group. T is finite; $A \sim \mathbb{Z}^r$;

Consequence: If A_1, A_2 abelian f.g. groups, then

$$\begin{array}{ccc} A_1 & \underset{c.e.}{\sim} & A_2 \\ \wr & & \wr \\ \mathbb{Z}^n & \sim & \mathbb{Z}^m \end{array} \Rightarrow n = m \qquad \Longleftrightarrow \quad rk(A_1) = rk(A_2).$$

c) $\mathbb{N} \underset{c.e.}{\sim} [0,\infty)$;



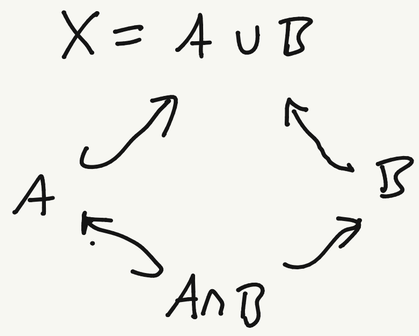
$$HX^*(\mathbb{N}) \cong HX^*([0,\infty)) \overset{\text{thm}}{\cong}$$

$[0,\infty)$: unif. contractible at infinity

$$HX^*(\mathbb{N}) \cong HX^*([0,\infty)) = \operatorname*{colim}_{n\to\infty} \overline{H}^{*-1}(X \setminus B(0,n)) = \overline{H}^{*-1}(*) = 0 \quad , *\in\mathbb{Z}$$

Mayer-Vietoris sequence

HX_* : Coarse $\longrightarrow$ $Gr_{\mathbb{Z}}$ Ab : (homology)



Paul D. Mitchener "Coarse homology theories"

Eilenberg–Steenrod axioms for coarse spaces

homology theory of.

$(h_n : Top \to Ab)_{n\in\mathbb{Z}}$;

$(hx_n : Coarse \to Ab)_{n\in\mathbb{Z}}$

$A, B \subseteq X$; (A,B) is excisive if $\forall E\in\mathcal{E}$, $\exists F\in\mathcal{E}$ st.

$$\underbrace{E[A] \cap E[B]}_{} \subseteq F[A\cap B]$$

Products in Coarse Cohomology

$(X,\mathcal{E})$ coarse space ; R a coefficient ring ;

$$HX^*(X) = HX^*(X;R)$$

$CX^*(X;R) \hookrightarrow (C^*(X;R), \cup)$

$\cup : C^n(X) \otimes C^m(X) \longrightarrow C^{n+m}(X)$

$$(\varphi, \psi) \mapsto (\varphi\cup\psi)(x_0,\dots,x_{n+m}) =$$

$$= \varphi(x_0,\dots,x_n)\,\psi(x_n,\dots,x_{n+m})$$

$\varphi\in CX^n$, $\psi\in CX^m \Longrightarrow \varphi\cup\psi \in CX^{n+m}$

$(CX^*(X), \cup) \rightsquigarrow (HX^*(X), \cup)$

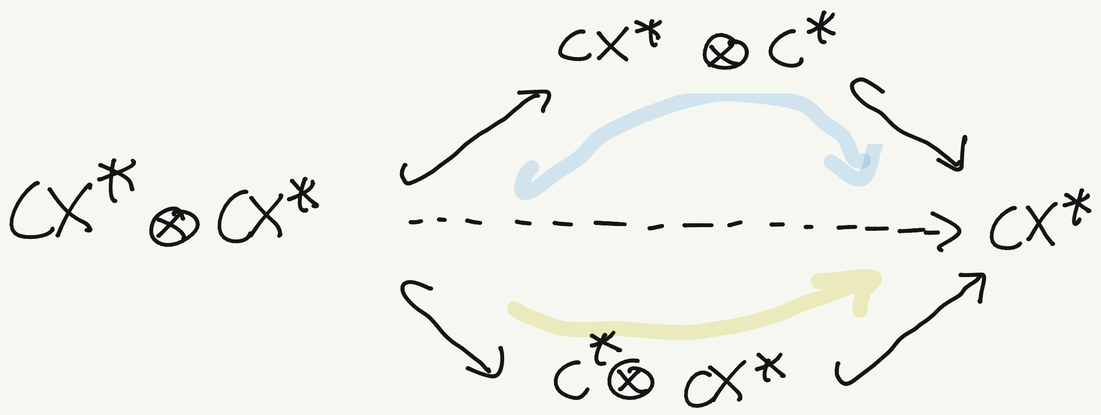
D.G.A. Graded (cohomology) algebra

$\alpha\cup\beta = 0$ in cohomology

$\cup$ vanishes in cohomology $CX^*(X) \trianglelefteq (C^*(X), \cup)$;

If $\varphi\in CX^*$, $\psi\in C^*$; $\varphi\cup\psi\in CX^*$

$\varphi\in C^*$, $\psi\in CX^*$; $\varphi\cup\psi\in CX^*$



$CX^* \triangleleft C^*$

Let $b \in X$ basepoint; $s: C^* \to C^{*-1}$;

$\varphi: X^{n+1} \to R \mapsto (s\varphi)(x_0, \cdots, x_{n-1}) = \varphi(b, x_0, \cdots, x_{n-1})$

$$CX^k \otimes CX^l \longrightarrow CX^{k+l-1}$$

$$\varphi * \psi := \underbrace{s\varphi}_{k-1} \smile \underbrace{\psi}_{l} + (-1)^k \underbrace{\varphi}_{k} \smile \underbrace{s\psi}_{l-1}$$

$$HX^k \otimes HX^l \xrightarrow{*} HX^{k+l-1}$$

"Roe product in Coarse Cohomology".

This Roe product in cohomology does depend on the choice of b, for $k, l \geq 2$.

- **Thm** (R. 2025): If (X,d) is a metric space, R coeff. ring;

  $\Sigma : HX^*(X) \longrightarrow HB^*(X)$ is a morphism of algebras
  $\quad *\qquad\qquad\quad *$

- **Corollary** (Computation of the Roe product):
  If X is unif. contr. at infinity, then

$$(HX^\bullet(X;R), *) \overset{(*)\ f}{\cong} \left(\operatorname{colim}_{B\in\mathcal{B}} \overline{H}^{*-1}(X\setminus B;R),\ \smile \right)$$

cup product.

$(*)$ $f(\alpha * \beta) = (-1)^{|\alpha|} f(\alpha) \smile f(\beta)$

- Brown representability theorem:

$$\text{Homology Theories}/\cong \quad \longleftrightarrow \quad \text{Spectra}/\sim \qquad \text{Top; CW-complexes.}$$

Coarse homology theories$/\cong$ $\longleftrightarrow$ Coarse spectra?$/\sim$

Ulrich Bunke & Alexander Engel
"Homotopy Theory w/ bornological coarse spaces"
Coarse Spectra using ∞-categories.

$$H_{lf} : \mathrm{Top} \to \mathrm{Ab}_{\mathbb{Z}} \quad \rightsquigarrow \text{limit process} \quad HX_{lf} : \mathrm{Coarse} \to \mathrm{Ab}_{\mathbb{Z}}$$